Practical Deployment Architecture

DSCI-41: Traffic Volume Estimation and Short-Term Congestion Forecasting

This document details the practical, real-world deployment strategy for our traffic forecasting system. It directly addresses how the AI model transitions from a research environment into a functional Intelligent Transportation System (ITS) capable of alerting the public and managing city-wide traffic flow.

> *Note on Scope: This application acts as ONLY a data provider. It does not attempt to replace existing consumer navigation apps. Instead, it generates highly accurate, pre-emptive congestion data that can be consumed by various endpoints.*

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1. Where and How Warnings are Issued

In a real-world municipal deployment, this JSON payload is distributed to four primary endpoints to ensure drivers receive timely warnings.

A. Variable Message Signs (VMS) on Highways

- Mechanism: The API integrates directly with the control software of physical VMS digital boards placed on highways.
- Implementation: If the model predicts "SEVERE" congestion at a roundabout in 15 minutes, the system sends an automated payload to the VMS located 5 kilometers before the bottleneck.
- End-User Experience: Approaching drivers see: "HEAVY CONGESTION 5KM AHEAD - USE SERVICE ROAD" allowing them to diverge before getting trapped in gridlock.

B. Third-Party Navigation Apps (Waze / Google Maps)

- Mechanism: Direct integration via the Waze for Cities (formerly Connected Citizens Program) data exchange.
- Implementation: Municipal traffic authorities partner with Waze. Our system feeds the predictive JSON data directly into the Waze ecosystem.
- End-User Experience: Because our model predicts congestion before it physically forms, Waze can pre-emptively reroute drivers to alternative paths, effectively distributing the traffic load and preventing the predicted jam from ever reaching critical capacity.

C. Centralized Traffic Command Dashboard

- Mechanism: A web-based Geographic Information System (GIS) dashboard built using React or Streamlit for human operators (Traffic Police).

- Implementation: The map displays roads that change color based on future predictions, not just current status.
- End-User Experience: If an anomaly is predicted (e.g., sudden heavy delay due to rain), the dashboard triggers a flashing red alert. A human operator can then dispatch physical traffic police to the intersection to manually manage flow before gridlock occurs.

D. Smart City Commuter Push Notifications

- Mechanism: Integration with local "Smart City" mobile applications via cloud notification services (e.g., AWS SNS, Firebase Cloud Messaging).
- Implementation: Commuters subscribe to alerts for their daily routes. If a predicted bottleneck intersects with their route, an automated push notification is dispatched.
- End-User Experience: The user receives an SMS/Notification: "Alert: Your usual route via NH-16 will experience heavy delays in 15 mins due to rain. Suggested alternate route: Main Street."

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2. Google Maps Integration Strategy

A common question regarding practical deployment is: "Can we push our data directly into Google Maps?"

The Strategy: We cannot directly alter the public Google Maps consumer app independently. However, the system scales at an enterprise level through the following authorized channels:

1. Authorized Municipal Partnerships (Waze for Cities): Waze acts as the ingestion point for municipal traffic data, which shares its ecosystem with Google Maps. This is the standard method used by actual city governments.
2. Custom API Overlays: For dedicated municipal use, we use the Google Maps Platform API. We load the base Google Map into our own application and programmatically draw our custom "red lines" (predicted congestion) on top of it.
3. OpenStreetMap (OSM) Routing: As a fully open-source alternative, the system can integrate with OSRM (Open Source Routing Machine). This takes our AI congestion predictions and independently calculates the fastest routes without relying on Google's proprietary ecosystem.

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3. Conclusion of Deployment Feasibility

By separating the Prediction Engine (our AI model) from the Dissemination Engine (VMS, Waze, Dashboards), the architecture remains highly modular, scalable, and identical to how enterprise ITS systems are deployed in modern smart cities.